

CMSC 25300 / 35300 (Fall 2024)

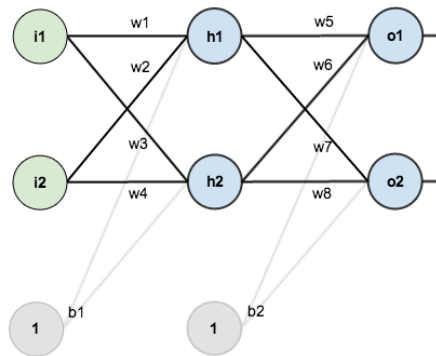
Homework 7

Submission Instructions:

- Please submit your homework in PDF to Gradescope (which can be accessed from the course's website on Canvas). Make sure to match all questions to the correct pages in Gradescope - points will be deducted if these are not properly matched.
- Note that you do not need to copy the problem statements in your solution, *as long as you clearly indicate the problem numbers* (e.g., 1.a, 2.c, etc).

1. Read <https://mattmazur.com/2015/03/17/a-step-by-step-backpropagation-example/>.

2. Consider this network architecture:



- a) (10 points) For this network architecture, what are the dimensions of the input i , the hidden layer h , and the output o ?
- b) (20 points) Now let's practice neural network's computation. Imagine we obtained the following networks with associated weight and bias parameters:
- $w1 = 0.06, w2 = 0.08, w3 = 0.10, w4 = 0.12$
 - $w5 = 0.14, w6 = 0.16, w7 = 0.18, w8 = 0.20$
 - $b1 = 0.22, b2 = 0.24$

As in the reading, assume the non-linearity is a sigmoid function:

$$\sigma(x) = \frac{1}{1 + e^{-x}}.$$

As in the reading, this non-linearity is applied when computing the hidden state values $h1, h2$ and the output states $o1, o2$.

If we input $i1 = 0.02$ and $i2 = 0.04$, what will outputs $o1$ and $o2$ be? Please show all calculations.

- c) (14 points) Assume the training labels are .01 and .99 for $o1$ and $o2$, respectively. What is the mean squared error L of the outputs of the forward pass? Please show all calculations.
- d) (48 points) What is the value of dL/dw_k for $k = 1, \dots, 8$? Please show all steps and calculations for each w_k .
- e) (8 points) Give an expression for the update of w_k in terms of a step size τ .